Yashraj Singh Thakur

+91 7000501987 Mail LinkedIn Github Portfolio Leetcode

Education

Narmada Valley International School, Narmadapuram **2019 – 2020**
Class 10th *92.8%*

Narmada Valley International School, Narmadapuram **2021 – 2022**
Class 12th *97.4%*

Indian Institute of Technology, Patna **2023 – 2027**
B.Tech in Engineering Physics *8.36/10 CGPA*

Work Experience

AlgoUniversity (backed by Y-Combinator) **May 2025 – June 2025**
SWE Co-Op Intern *Remote*

- Built an automated code evaluation platform supporting **3+ languages** and handling over **100 concurrent users**.
- Analyzed execution logs to identify performance bottlenecks and reduced average latency by **25%**.
- Implemented authentication workflows to manage user access and progress tracking for 40+ users.
- Deployed containerized services to standardize runtime environments and improve system stability.

Projects

Cohort – Online Competitive Programming Judge GitHub Deployment

- Designed a queue-based task processing system to manage peak loads of **1,000+ submissions**.
- Monitored traffic patterns to identify system bottlenecks and improve processing throughput.
- Deployed cloud infrastructure achieving **99.9% uptime** for continuous service availability.
- Automated submission evaluation workflows to reduce manual intervention and execution delays.
- **Tools:** Node.js, MongoDB, Redis, AWS EC2, Vercel.

Algorithmic Trading Strategy GitHub

- Developed a forecasting-based decision system using time-series indicators for automated trade signals.
- Validated strategy behavior using historical data and automated testing.
- Containerized services to ensure reproducible deployments across environments.
- Measured system reliability using automated test coverage of **82%**.

Smart Energy Forecast Dashboard *(FastAPI, ML, RL, PostgreSQL)* GitHub Link

- Built a demand forecasting and resource scheduling system using historical consumption data.
- Reduced estimated energy cost by **15–20%** through optimized scheduling strategies.
- Tracked system performance using dashboards for cost and emission monitoring.
- Applied reinforcement learning to improve utilization of available energy resources.
- **Tech:** FastAPI, XGBoost, Prophet, PostgreSQL, Chart.js.

Technical Skills

Programming: Python, Java, JavaScript, C

Data & Systems: SQL, PostgreSQL, MongoDB, Redis, Docker, AWS, GCP

Frameworks: Node.js, Express.js, FastAPI, React

Tools: GitHub, Postman, VS Code